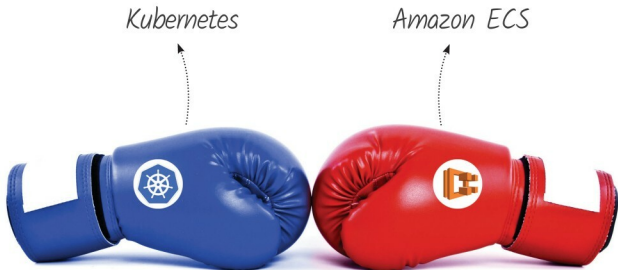


The Container Wars: Kubernetes vs Amazon ECS

Among container orchestration technologies, Kubernetes and Amazon ECS lead the pack. However, on what basis should systems administrators choose one over the other? This article will help them make an informed choice.



Shipping containers are regarded as ubiquitous and standardised. They can be accessible from anywhere in the world by following simple procedures. Software containers operate in somewhat a similar manner. Packing the software container involves defining what is required for your application to work. This includes the operating system, library files, configuration files, application binaries and other components of the technology stack.

A container image is used to create containers that can operate in almost any environment, from standard desktop systems to data centres and even on the cloud. Containers are regarded as highly efficient as many of them can run on the same machine allowing full resource utilisation. Container technology was introduced in the late 1970s but became extremely popular only when Docker was introduced in 2013. Since then, container technology has drastically changed the DevOps landscape and the way applications are developed, shipped and run.

A single container ordinarily runs a single application. The running of multiple operations simultaneously with a

single container has been made possible with the concept of container orchestration.

The importance of containers

- **Effective DevOps and continuous delivery:** When the application consists of multiple containers with clear interfaces among them, it is simple to update a container, assess the impact and even roll back to an earlier version. By having multiple containers, the same capability enables updating every container without any sort of hiccup.
- **Environment replication:** When making use of containers, it is fairly easy to instantiate identical copies of the full application stack and configuration. This can be performed by using new developers and new support teams to safely research and experiment in total isolation.
- **Testing:** With containers, a stress test can be performed to ensure the environment required matches with the normal operational requirements.
- **Scalability:** By building an application from multiple container instances, adding more containers scales out